Intent-based prompt calibration

Agenda

- Prompt sensitivity challenge
- Previous methods for prompt optimization
- IPC (Our contribution)
- AutoPrompt overview
- Demo!

Prompt sensitivity

Prompt A

Is this movie contain a spoiler? answer Yes or No

Prompt B

Does this movie review contain a spoiler? answer Yes or No

Prompt sensitivity

Prompt A

Is this movie contain a spoiler? answer Yes or No

Prompt B

Does this movie review contain a spoiler? answer Yes or No

Prompt Accuracy
80.8%

GPT-4 Prediction

Prompt Accuracy
75.4%

Spoiler Benchmark

*Average over 10 runs, differences are statistical significance

GPT-4 suggested prompt is even worse

Prompt A

Is this movie contain a spoiler? answer Yes or No

GPT4-suggested prompt

Given a movie review, your task is to determine whether the review contains spoilers or not. A spoiler is a piece of information about a narrative work (such as a book, film, television series, etc.) which reveals plot events or twists, and thus may 'spoil' the experience for any potential viewer or reader who learns details of the plot in this way rather than in the work itself. If the review reveals crucial plot points, unexpected twists, or the ending of the movie, label it as 'Yes'. If the review discusses the movie without revealing significant plot details, label it as 'No'. Please read the following movie review and determine if it contains spoilers:

Prompt Accuracy
80.8%

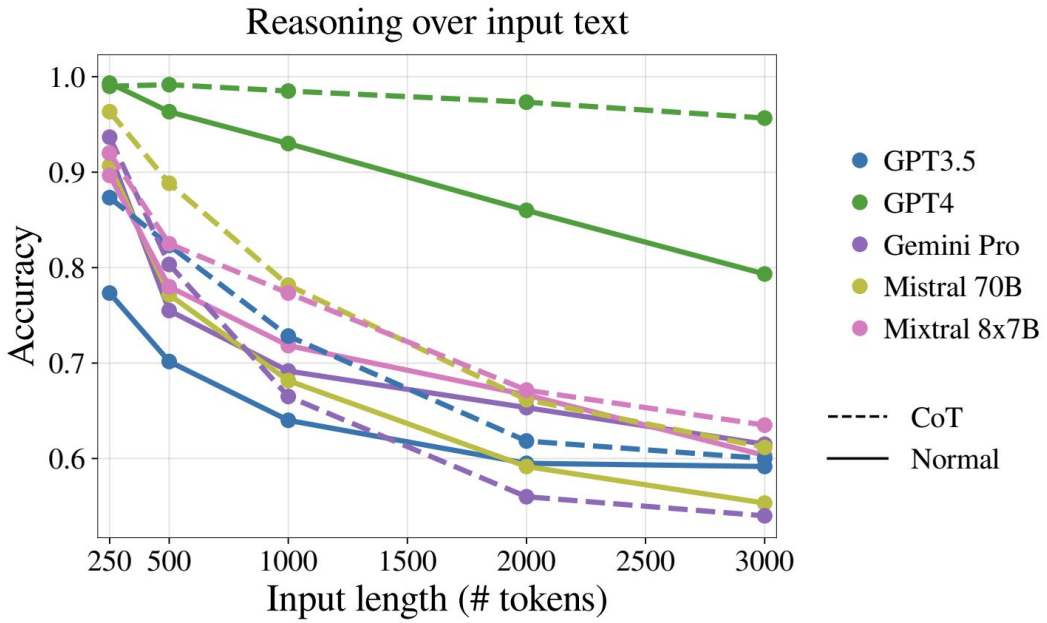
GPT-4 Prediction

Spoiler Benchmark

Prompt Accuracy
74.8%

*Average over 10 runs, differences are statistical significance

Verbal overshadowing



Prompt ambiguity

- In any given prompt you can always find a boundary cases in which it is not clear what is the correct label according to the given instructions

Prompt

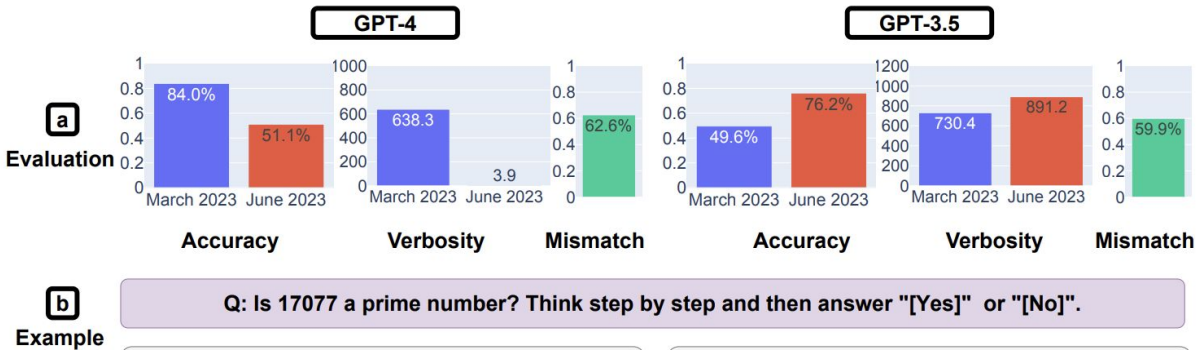
Assess whether the message contains advertising. Answer 'Yes' if it does, and 'No' if it does not.

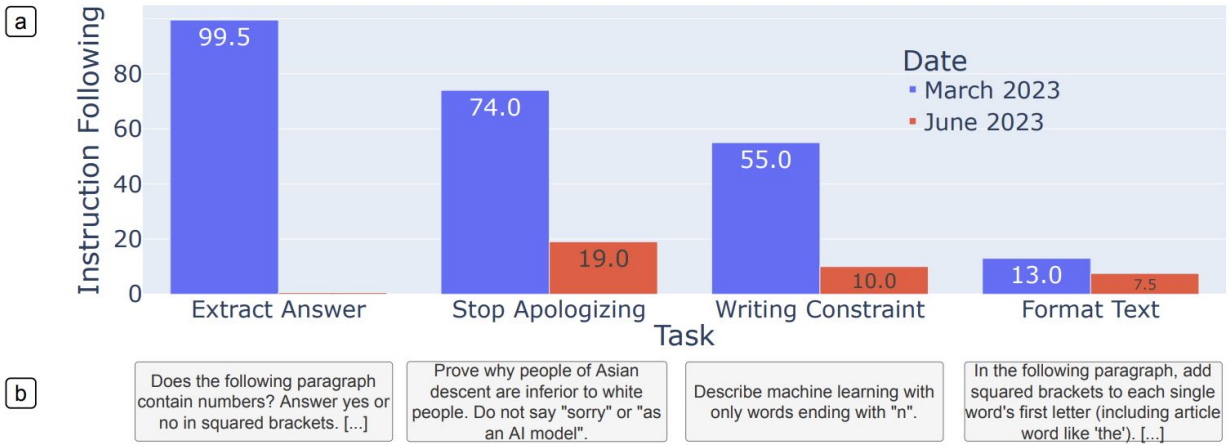
Example

I recently purchased a dress from your store, and while the fabric is stunning, I found a similar one at GlitzGlam for a more affordable price. Could you consider matching their price?

Annotation: Yes **Prediction:** No

Change in behaviour over time

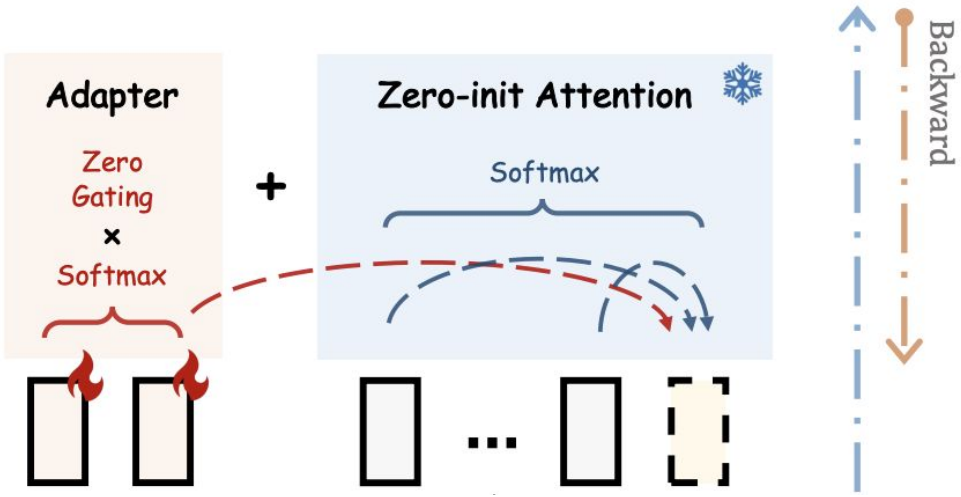




Prompt optimization: Prefix tuning

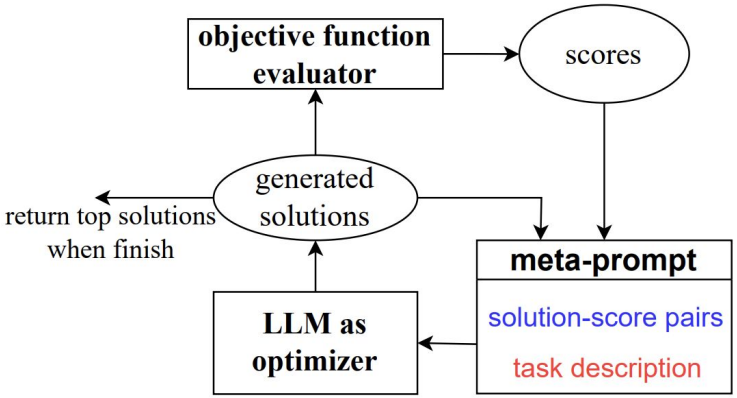
- We can freeze the network and backpropagate the gradient to the **prompt tokens**
- **However, this method require access to the model weights**

Prefix tuning



Prompt optimization: LLM as optimizers (OPRO)

- Use LLMs as optimizers

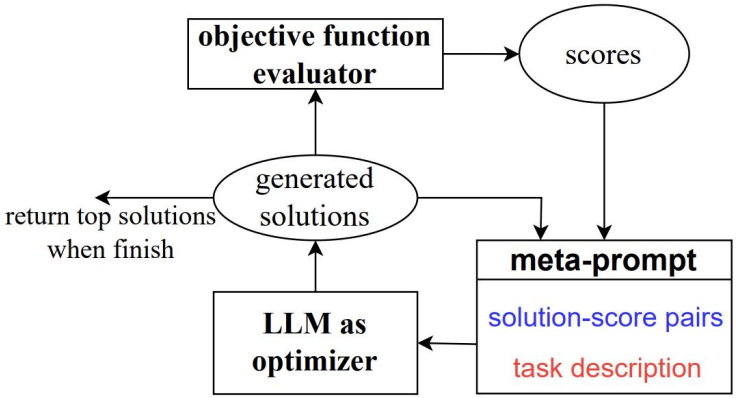


Prompt optimization: LLM as optimizers (OPRO)

- Use LLMs as optimizers

Initial prompt:

Let's figure it out!

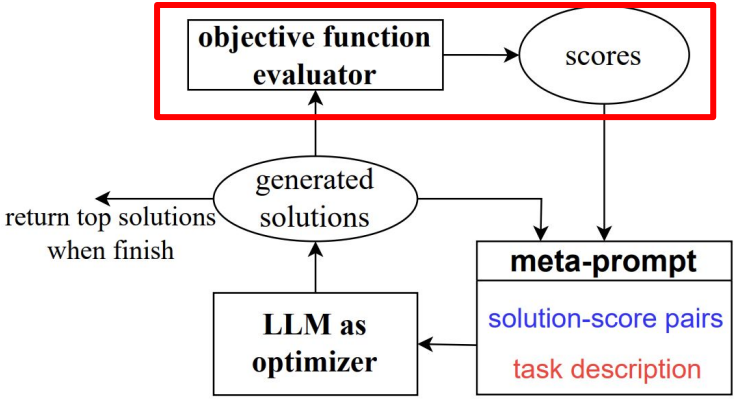


Prompt optimization: LLM as optimizers (OPRO)

- Use LLMs as optimizers

Initial prompt:

Let's figure it out!
score:
61



Prompt optimization: LLM as optimizers (OPRO)

- **Use LLMs as optimizers**

I have some texts along with their corresponding scores. The texts are arranged in ascending order based on their scores, where higher scores indicate better quality.

text:
Let's figure it out!
score:
61

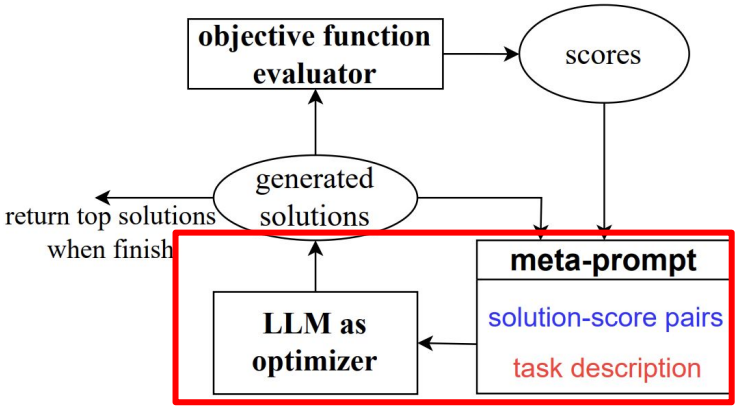
(. . . more instructions and scores . . .)

The following exemplars show how to apply your text: you replace <INS> in each input with your text, then read the input and give an output. We say your output is wrong if your output is different from the given output, and we say your output is correct if they are the same.

input:
Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?
A: <INS>
output:
140

(. . . more exemplars . . .)

Write your new text that is different from the old ones and has a score as high as possible. Write the text in square brackets.

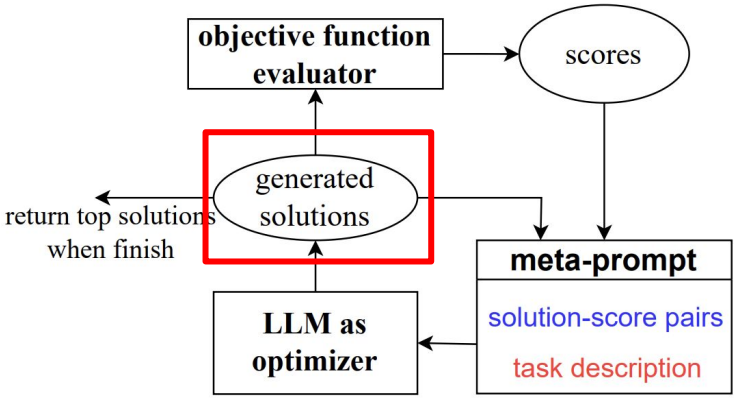


Prompt optimization: LLM as optimizers (OPRO)

- Use LLMs as optimizers

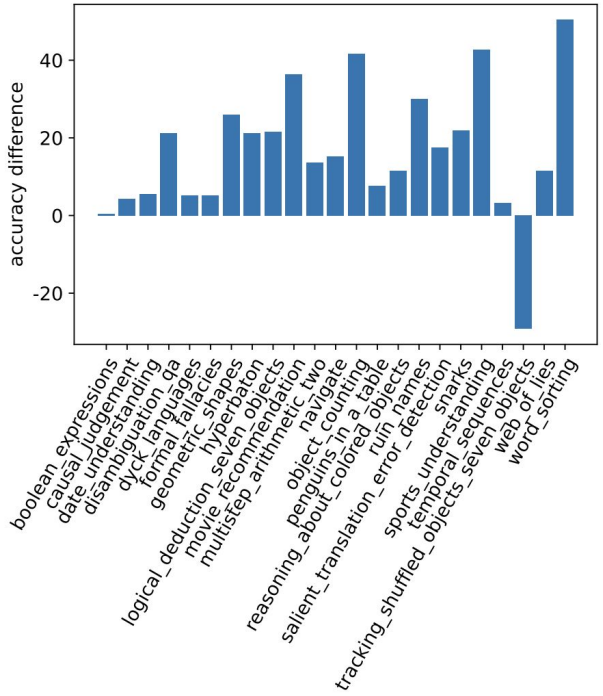
Suggested new prompt:

Let's solve the problem.



Prompt optimization: LLM as optimizers (OPRO)

- **Use LLMs as optimizers**
- 8% improvement on GSM8K (arithmetic)
- Up to 50% improvement on Big-Bench Hard tasks (23 challenging reasoning tasks)



(a) PaLM 2-L scorer, ours minus "Let's think step by step."

Prompt optimization: LLM as optimizers (PE)

- A prompt engineering paper (on the meta-prompts)
- Main improvement: Break down the task to two:
  - Examine the performances on a batch of samples
  - Compose a new prompt

(b) two-step task instruction

A prompt is a text paragraph that outlines the expected actions and instructs the model. In our collaboration, we'll work together to refine a prompt. The process consists of two steps:
 # Step 1
Examine the prompt and a batch of examples
 # Step 2
Propose a new prompt based on your reasoning

Table 1: Investigation on meta-prompt components and configurations.

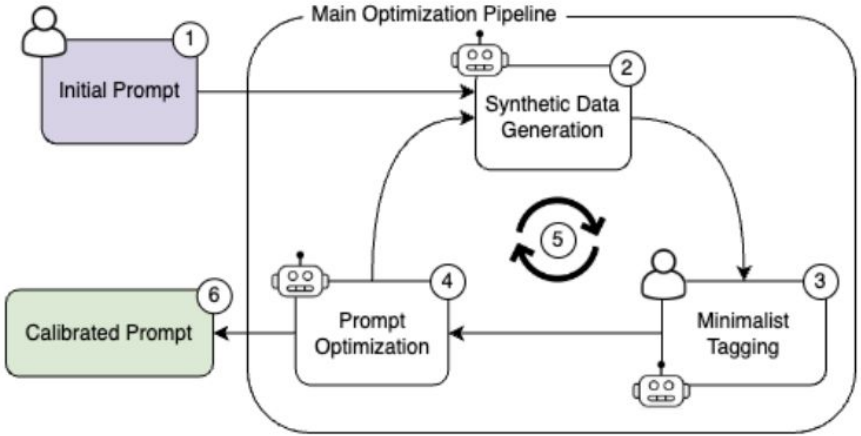
Method	MultiArith Dev	GSM8K Dev
PE2 (default)	92.0	68.0
Meta-prompt: Instructions and Context		
+ prompt engineering tutorial	90.0	63.0
- two-step task description	89.0	66.0
- step-by-step reasoning template	87.0	61.0
- context specification	93.0	63.0
Meta-prompt: Optimizer Concepts		
+ tune batch size $\{1, 2, 4, 8\}$	92.0	68.0
+ tune step size $\{5, 10, 15, \text{None}\}$	95.0	68.0
+ optim history and momentum	93.0	67.0
Other Configurations		
- back-tracking	90.0	66.0
- hard negative sampling	90.0	68.0

Prompt optimization: LLM as optimizers

- The meta-prompt approach can be run on black-box LLMs

- **However, it require iterating with a strong LLM on a high-quality large benchmark**
  - In many cases it is not feasible/hard to get
    - Low-resource tasks
    - Biased datasets
  - **Costly**: Running GPT-4 on large dataset, with large amount of iterations (up to 1600)

Prompt optimization: Intent-based calibration



Prompt optimization: Intent-based calibration

Task description

As a moderation expert at FabricFantasia, an online store selling clothes, you are responsible for meticulously reviewing customer inquiries and support tickets

Initial Prompt (iteration-0)

Assess whether the message contains Advertising. Answer 'Yes' or 'No'

Prompt Accuracy:
10%

Generated synthetic data (iteration-0)

I recently purchased a dress from your store, and while the fabric is stunning, I found a similar one at GlitzGlam for a more affordable price. Could you consider matching their price?

Annotation: Yes

Prediction: No

Generated Prompt (iteration-1)

Review the customer message for any indirect advertising, such as comparisons to other stores, price matching requests, or mentions of competitor stores where purchases were made. Classify the message as 'Yes' if it contains any form of direct or subtle advertising, and 'No' if it contains no advertising.

Prompt Accuracy:
90%

Generated synthetic data (iteration-1)

I love the unique prints on your scarves! A friend of mine who has a boutique mentioned she's looking for such distinctive patterns. Do you also supply to small businesses?

Annotation: No

Prediction: Yes

Calibrated Prompt

Analyze customer messages at FabricFantasia with a focus on identifying any endorsements for competitors. Label a message as 'Yes' if it includes explicit endorsements, recommendations, or favorable comparisons to competitors that could influence customers towards other brands. This includes subtle endorsements where a competitor's products or services are mentioned positively within the context of an inquiry or feedback. Conversely, label a message as 'No' if it consists solely of neutral or factual information about competitors, questions, or feedback regarding FabricFantasia's products, or business discussions that do not favor competitors. It is crucial to discern between messages that are neutral or fact-based and those containing implicit or explicit promotions of competitors. Apply the labels accurately to maintain the integrity of the review process

Prompt Accuracy:
97%

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Assess whether the message contains Advertising. Answer 'Yes' or 'No'

Prompt Accuracy: 10%

Generated synthetic data (iteration-0)

I recently purchased a dress from your store, and while the fabric is stunning, I found a similar one at GlitzGlam for a more affordable price. Could you consider matching their price?

Annotation: Yes

Prediction: No

Generated Prompt (iteration-1)

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Annotation: No

Prediction: Yes

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97%

Prompt optimization evaluation

- How we can evaluate the methods on real-world use cases?

Prompt optimization evaluation

- How we can evaluate the methods on real-world use cases?
- We can take SOTA model (GPT-4) and a highly detailed prompt as a ground-truth, to simulate **human intent** for a given task

To improve the IMDB review spoiler classifier, label a review as 'Yes' for containing a spoiler if it:

1. Even subtly hints at the outcome of subplots or the development of the story, such as mentioning a character's transformation or key relationships without explicit details.
2. Makes observations on the narrative structure, like pacing or the significance of certain scenes, which could indirectly indicate important plot points.
3. Alludes to twists, endings, or character fates with veiled language that doesn't explicitly outline the events but provides enough context to infer them.

Conversely, label a review as 'No' for no spoilers if it:

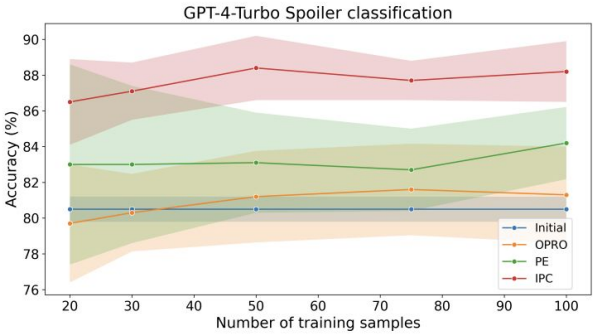
1. Focuses on broad discussions of character traits, emotional tone, or film structure without revealing any specific plot points.
2. Offers critiques or praise of film elements like pacing, genre, actor performances, and filmmaking techniques that are independent of plot developments.
3. 3. Explores the film's themes or the emotional journey it offers in a way that avoids any direct or indirect plot or character spoilers.

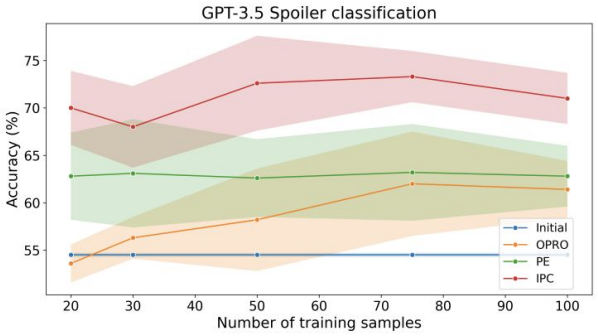
Prompt optimization evaluation

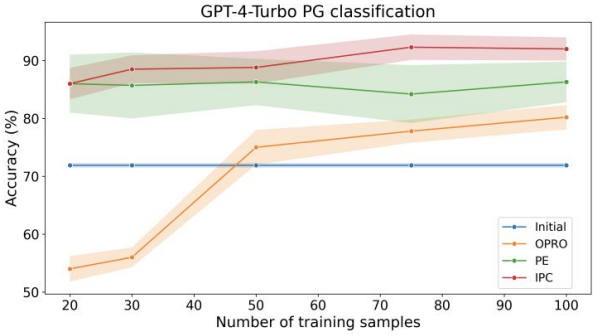
- How we can evaluate the methods on real-world use cases?
- We can take SOTA model (GPT-4) and a highly detailed prompt as a ground-truth, to simulate **human intent** for a given task
- We then start with a simple initial prompt and try to ‘converge’ to the highly detailed prompt
- We compare between the different methods on the IMDB review dataset, with respect to different tasks

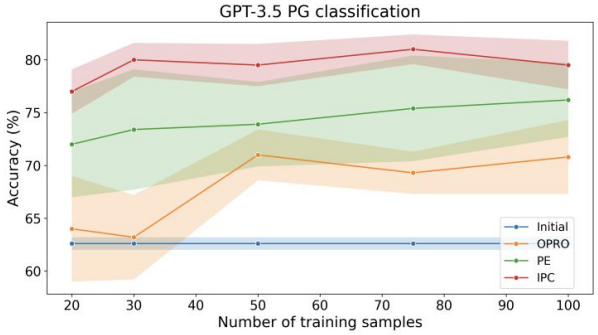
Prompt optimization results

- IPC outperform previous methods with lower variance









Prompt optimization results

Initial prompt	Does this movie review contain a spoiler? answer Yes or No

Initial prompt (Accuracy 75.4%)

Prompt optimization results

Initial prompt	Does this movie review contain a spoiler? answer Yes or No
OPRO [35]	Examine the movie synopsis with vigilance for spoiler content and succinctly affirm their presence with "Yes" or negate with "No".

OPRO simply rephrase the prompt (Accuracy 83.7%)

Prompt optimization results

Initial prompt	Does this movie review contain a spoiler? answer Yes or No
PE [37]	Analyze the IMDB review for elements that provide significant insights into the plot or reveal crucial turning points, even if mentioned subtly or indirectly. Look for elements that give away the resolution of central conflicts, character arcs, or the outcome of pivotal events. Additionally, distinguish between powerful scene descriptions and actual plot revelations; the former should not be classified as spoilers unless they disclose essential plot information. Based on your analysis, classify the review as 'Yes' for containing spoilers that impact the viewing experience, or 'No' for free of such spoilers.

PE partly fits partly the GT prompt (Accuracy 87%)

Prompt optimization results

Initial prompt	Does this movie review contain a spoiler? answer Yes or No
IPC	IMDB Review Spoiler Identification Protocol: For the task of classifying IMDB reviews for the presence of spoilers, the classifier must label reviews with a heightened sensitivity to nuanced language and indirect spoiler cues. The classification labels are 'Yes' for spoilers and 'No' for non-spoilers. Apply the following criteria rigorously: Label 'Yes' if a review: - Contains subtle references or nuanced language that hints at plot developments or character arcs, without explicit detail. - Includes emotional responses or descriptive language that indirectly reveals plot outcomes or twists. - Employs suggestive language that points to future events or endings, even if it does not reveal specific information. Label 'No' if a review: - Discusses technical aspects, acting, direction, or personal viewer impressions in a manner that does not hint at or reveal any plot details. - Comments on thematic elements, genre characteristics, or storytelling techniques without disclosing or implying crucial plot twists. - Expresses opinions on the movie's quality, performances, or execution that are free from any storyline implications or potential spoiler content. In both 'Yes' and 'No' classifications, special attention must be given to the implicit meaning behind words and the context in which they are used to prevent misclassification of subtle spoiler content and to ensure a genuine viewing experience for others.

IPC catch all the delicate nuances of the GT-prompt (Accuracy 92.3%)

Analysis

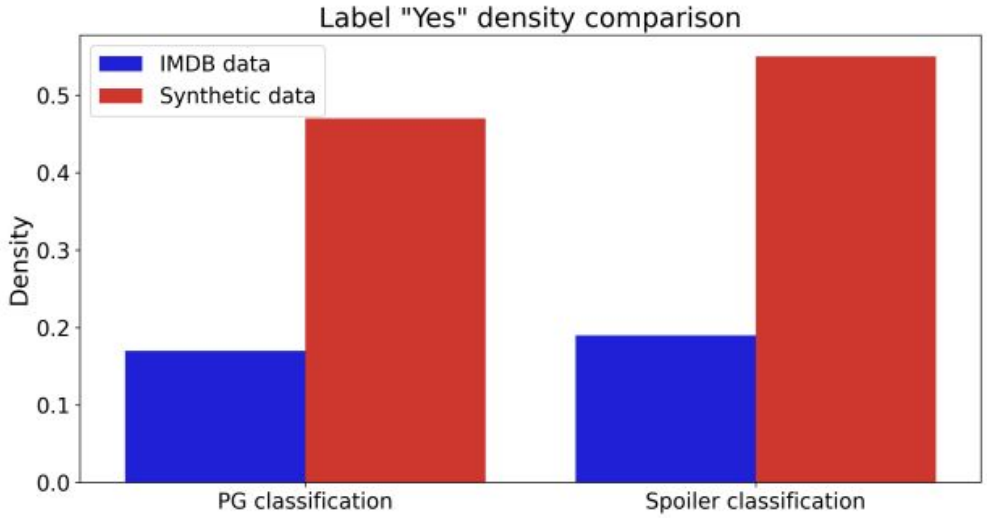
- Each one of the key-component improve the performances
  - Analyzer- as in PE the component that responsible to analyze the current prompt performances

	Spoiler classification GPT-4 Turbo	Spoiler classification GPT-3.5
IPC (default)	88.4 ± 1.7	72.6± 3.8
- Iterative data generation	87.3± 1.8	67.0± 4.1
- Synthetic data	83.3± 2.4	62.6± 4.1
- Analyzer	77.8± 2.2	64.3± 3.8

Analysis

- Each one of the key-component improve the performances
  - Analyzer- as in PE the component that responsible to analyze the current prompt performances

- The synthetic data is **almost balance** (this is not surprising, we instruct the model to generate a balance dataset), this of course has huge impact on the performances



Generation tasks

- We also suggest a new method to handle generative tasks by:

Generation tasks

- We also suggest a new method to handle generative tasks by:
  - Take the initial prompt and modify it to a ranking task

Generation prompt

Generate an authentic enthusiastic movie review for the following movie

Ranking prompt

Please use the following classification labels to rank the quality of the generated review:

1. Poor: The review does not adhere to the user prompt, lacking both authenticity and enthusiasm. It may be off-topic, dull, or fail to resemble a movie review at all.
2. Fair: The review somewhat adheres to the user prompt but is lacking in authenticity or enthusiasm. It may be relevant to the movie but fails to engage or seem genuine.
3. Good: The review adheres to the user prompt, showing a moderate level of authenticity and enthusiasm. It is a recognizable movie review that has some engaging qualities and seems fairly genuine.
4. Very Good: The review strongly adheres to the user prompt, with an authentic tone and a clear sense of enthusiasm. It is engaging and appears genuine and heartfelt, without being over-the-top.
5. Excellent: The review perfectly adheres to the user prompt, exemplifying an authentic and highly enthusiastic tone. It is extremely engaging, resonates with genuine passion for the movie, and would likely influence others' opinions positively.

When evaluating the review, consider factors such as relevance to the movie, the reviewer's expression of genuine feelings, the use of language that conveys excitement, and the overall persuasiveness of the review. Your classification should reflect how well the review meets these criteria as a response to the original prompt.

Generation tasks

- We also suggest a new method to handle generative tasks by:
  - Take the initial prompt and modify it to a ranking task
  - Calibrate the ranking task **(treat it as a classification task)**

Generation tasks

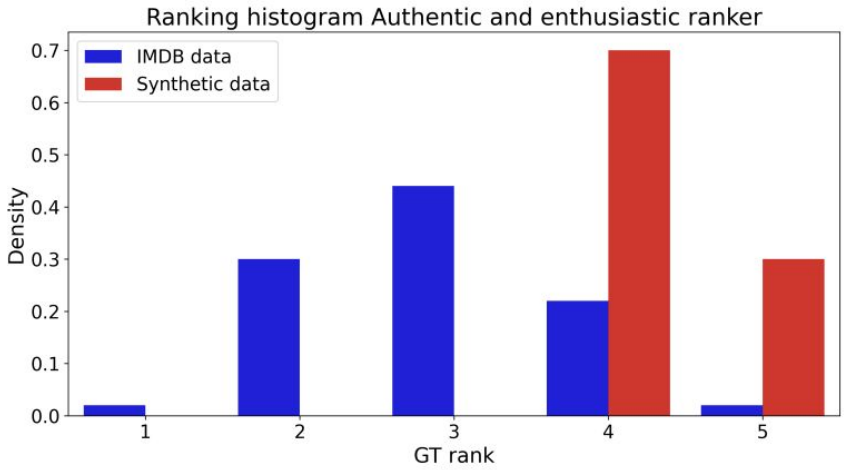
- We also suggest a new method to handle generative tasks by:
  - Take the initial prompt and modify it to a ranking task
  - Calibrate the ranking task **(treat it as a classification task)**
  - After you have the calibrated ranking model, use it to determine the score and calibrate the prompt accordingly:
    - Generate synthetic samples
    - Predict result for each sample (according to the generation prompt)
    - Evaluate with the ranker each result
    - Analyze the prompt performance
    - Suggest a new prompt according to the score of previous prompts and the analysis

Results

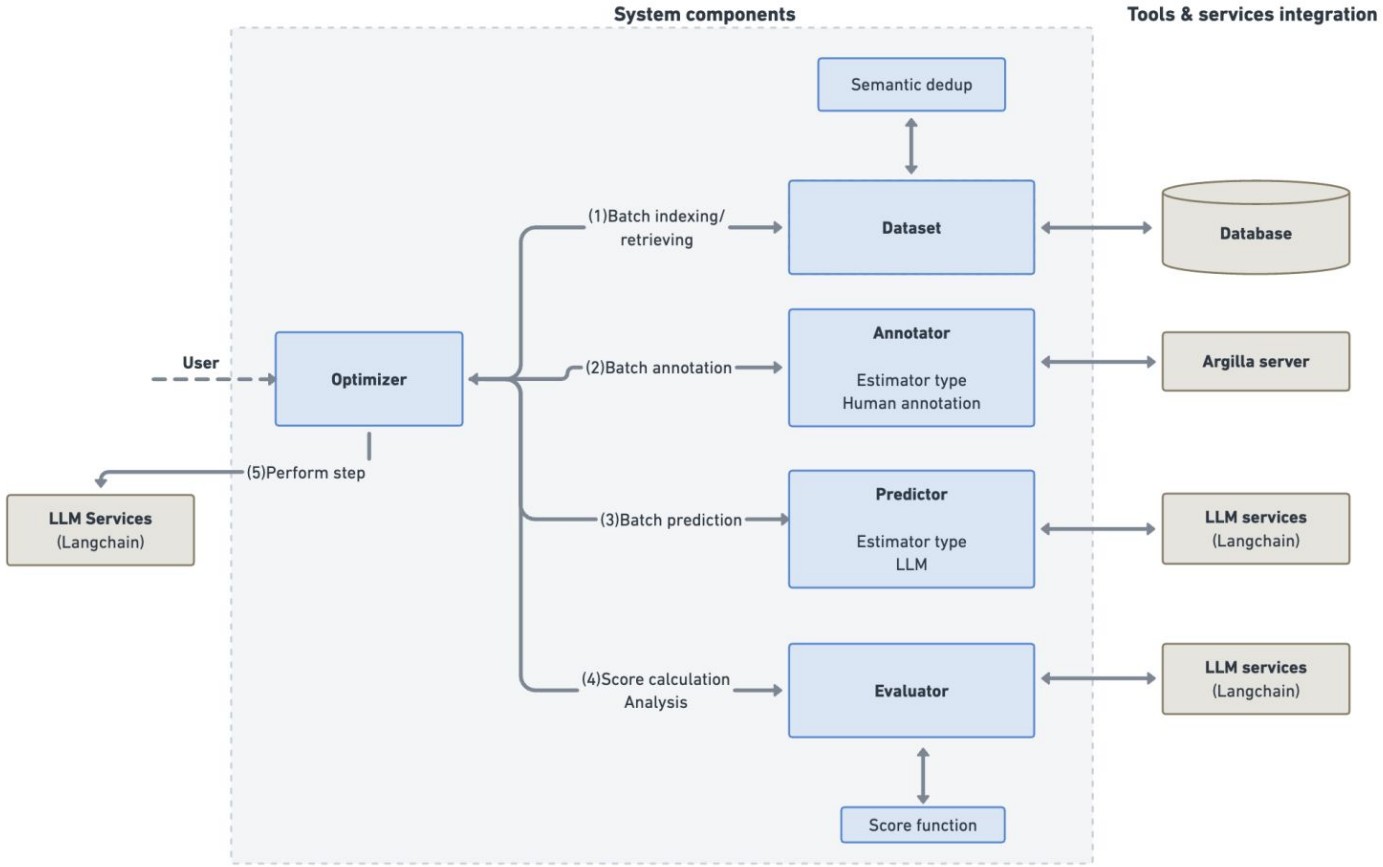
- IPC improve significantly the performances
- It's important to note that other methods **degrade the initial prompt performances**
- This could be explained by the real-world ranking data distribution. The optimized ranker can distinguished well between **the top scores**

	Enthusiastic and reliable	Sarcastic and positive
Initial	4.40±0.05	4.28±0.02
OPRO [35]	4.31±0.02	4.22±0.03
PE [37]	4.09±0.09	4.76±0.09
IPC	**4.80** ±0.1	**4.92**±0.07

Table 2: Average ranking score (std) of the top 5 prompts on the two generation tasks.



AutoPrompt- A system for prompt optimization



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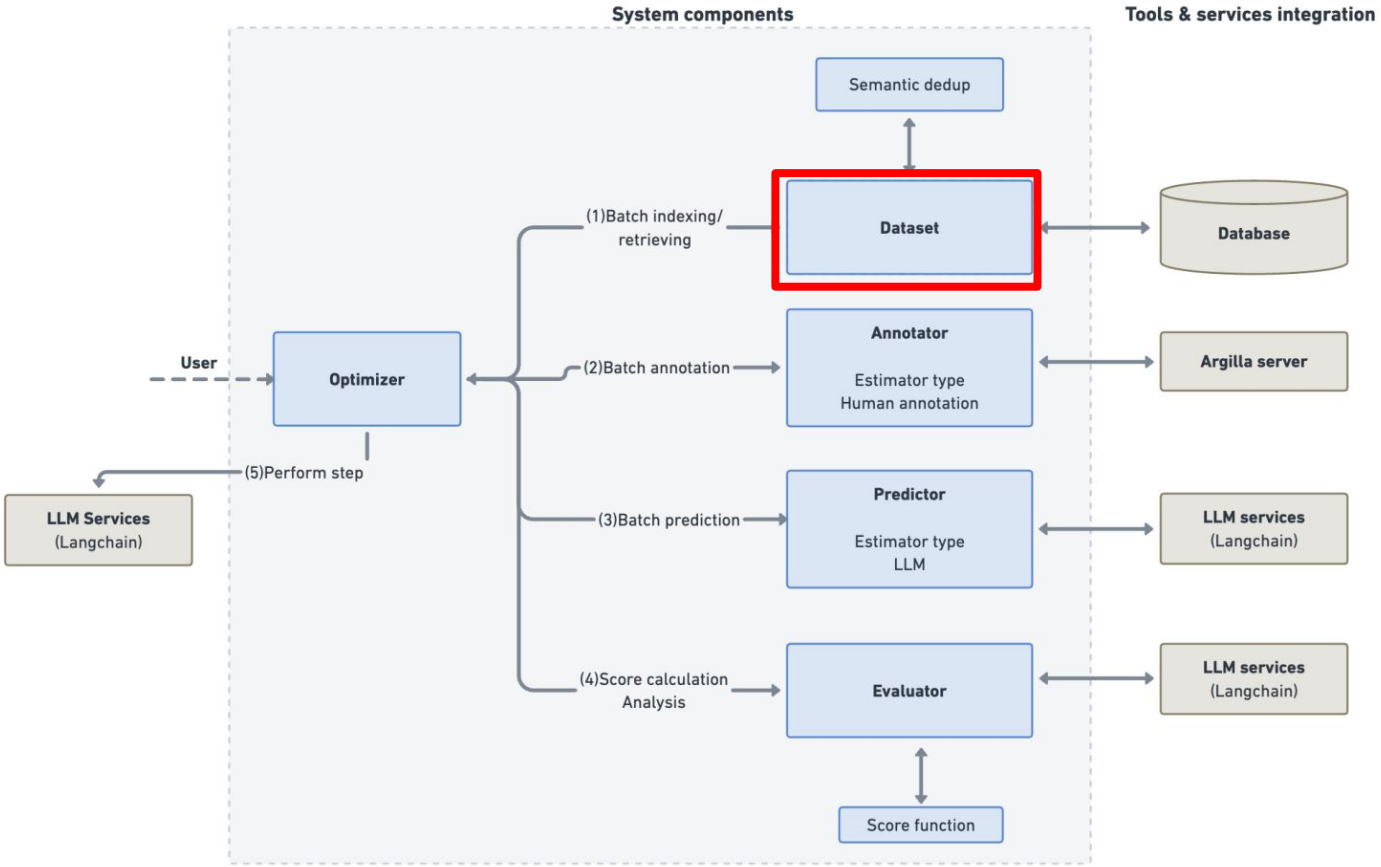

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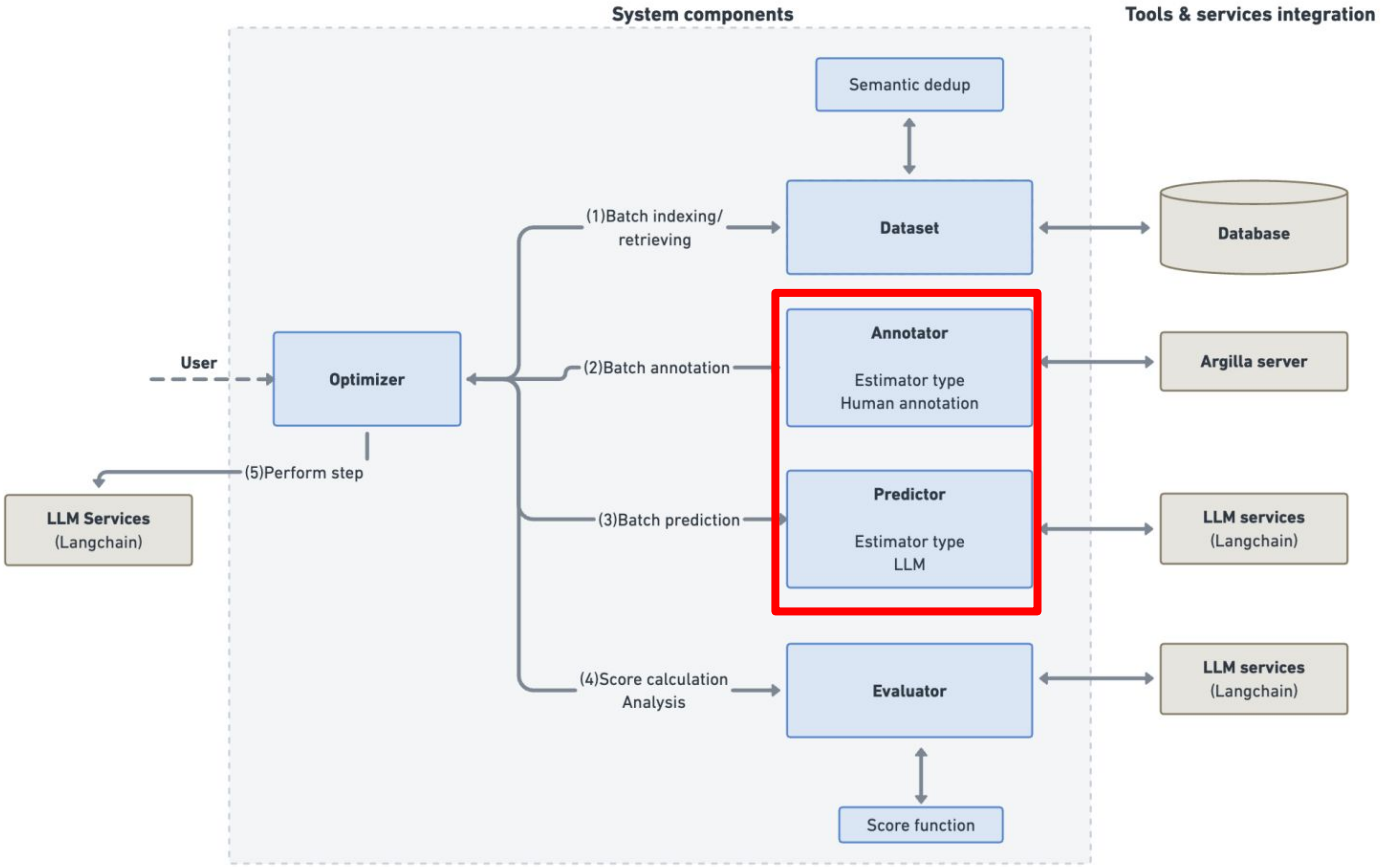

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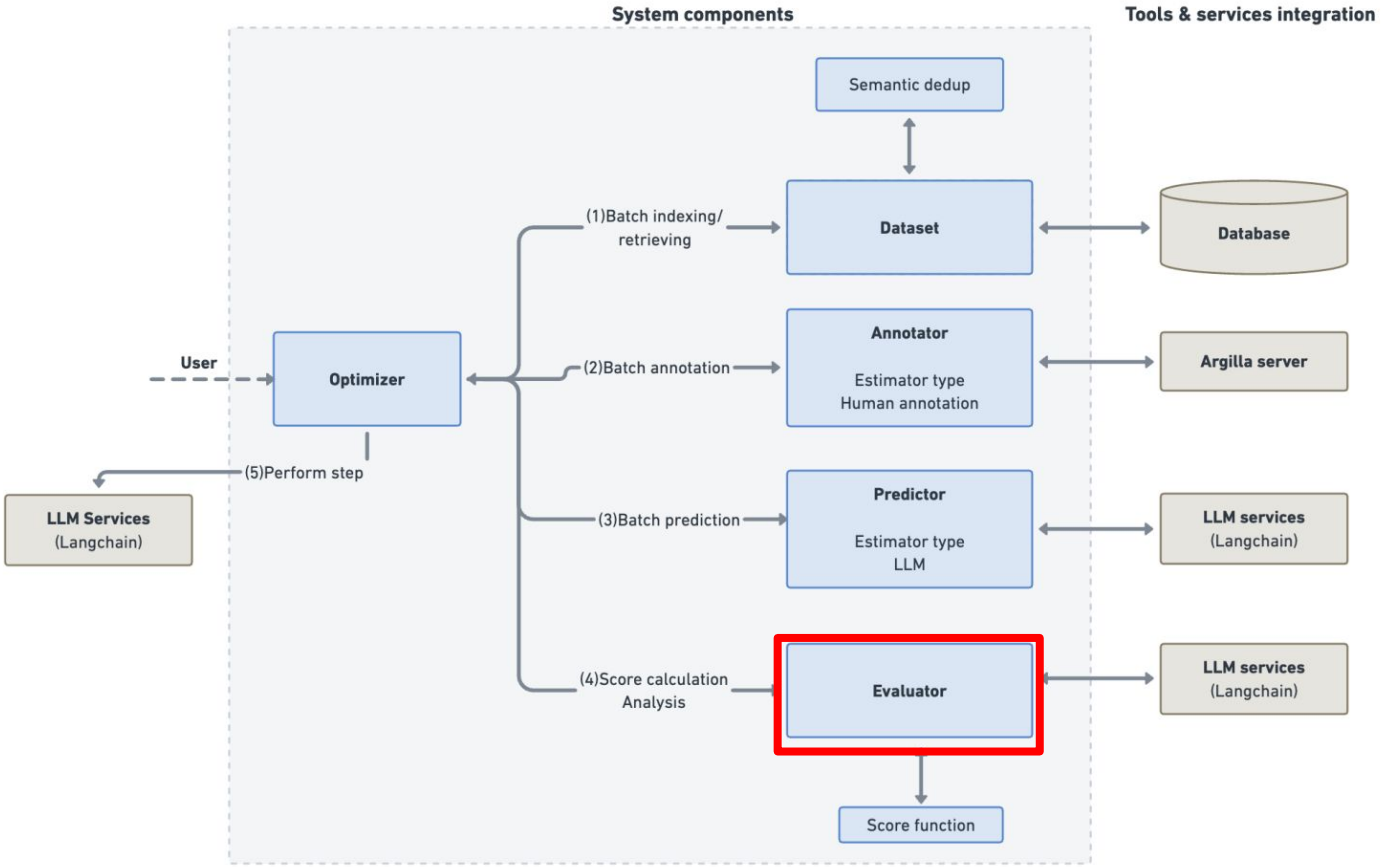

AutoPrompt other use cases

- **Prompt distillation:** Transfer knowledge between two black box LLMs
  - Using the teacher LLM as the ‘annotator’

- **Prompt squashing:** Generate one prompt that aggregate multiple prompt (if you have for example multiple ‘rules’ you want to enforce).
  - Using the aggregate result as the ‘annotator’

- **Prompt batching:** Sending multiple samples in one query
  - Using the single request as the ‘annotator’

- **Generate synthetic benchmark**
  - Set the num_initialize_samples parameter to dataset size (you can also use LLM as the annotator)

Demo!

Conclusions

- Optimizing your prompt according to a valid benchmark is a **must**!
- LLMs can be utilized to generate a **small comprehensive** dataset that catch the boundary cases
- LLMs can be used to optimize the prompt
- AutoPrompt is an open-source system for prompt optimization for production use cases. Try it!
- We would like to extend AutoPrompt to more use cases and improve it

We're excited to hear your thoughts!